# Huawei Access Node YEL-ACC-67 SOP for Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) Failure

**Node:** YEL-ACC-67 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_9857

## Description

This Failure Domain occurs on Huawei Access Node YEL-ACC-67 due to the inherent characteristics of the Routing Protocol Flap, which is a critical issue in the BGP, OSPF, and ISIS routing protocols. When the routing protocols experience a sudden and repeated change in their routing information, it can cause a 'flapping' effect, where the routing information is constantly being updated, resulting in a temporary loss of routing adjacency. This can be exacerbated by the complex interaction between the routing protocols, the underlying network topology, and the hardware capabilities of the Access Node. In the case of Huawei Access Node YEL-ACC-67, the issue is further compounded by the potential for hardware-related issues, such as faulty interface cards or memory corruption, which can cause the routing protocols to become unstable. To mitigate this issue, it is essential to perform a thorough analysis of the routing protocol configuration, network topology, and hardware configuration to identify the root cause of the problem. This SOP outlines the steps to be taken to diagnose and resolve the issue, ensuring minimal downtime and optimal network performance.

## Pre-requisite Checks

- Ensure remote hands are dispatched to the Yelahanka Sub-station to physically access the Huawei Access Node YEL-ACC-67.

- Verify the status of the redundant links to ensure they are functioning properly.

- Check the network topology to identify any potential bottlenecks or misconfigurations that may be contributing to the Routing Protocol Flap.

## Potential Root Causes

| Cause | Probability |
|---|---|
| BGP route churn due to misconfigured route filtering or incorrect routing policies. | High |
| OSPF adjacency loss due to incorrect interface configuration or misconfigured OSPF protocol parameters. | Medium |
| ISIS LSP flooding due to incorrect LSP configuration or misconfigured ISIS protocol parameters. | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display current-configuration | include bgp, Expected Output: BGP configuration details, including routing policies and filter lists., Description: This command will display the current BGP configuration, allowing us to identify any misconfigured route filtering or incorrect routing policies that may be contributing to
```

```
the BGP route churn.

Command: display ospf neighbor | include yelahanka-sub-station, Expected Output: OSPF neighbor details, including interface configuration and protocol parameters., Description: This command will display the current OSPF neighbor details, allowing us to identify any incorrect interface configuration or misconfigured OSPF protocol parameters that may be contributing to the OSPF adjacency loss.

Command: display isis lsp | include yelahanka-sub-station, Expected Output: ISIS LSP details, including LSP configuration and protocol parameters., Description: This command will display the current ISIS LSP details, allowing us to identify any incorrect LSP configuration or misconfigured ISIS protocol parameters that may be contributing to the ISIS LSP flooding.
```

## Resolution Steps

Step 1: Step: Correct BGP route filtering and routing policies., Command: undo bgp router-id yelahanka-sub-station, Description: This command will correct the BGP router ID and remove any misconfigured route filtering or incorrect routing policies that may be contributing to the BGP route churn.

Step 2: Step: Correct OSPF interface configuration and protocol parameters., Command: undo ospf router-id yelahanka-sub-station, Description: This command will correct the OSPF router ID and remove any incorrect interface configuration or misconfigured OSPF protocol parameters that may be contributing to the OSPF adjacency loss.

Step 3: Step: Correct ISIS LSP configuration and protocol parameters., Command: undo isis lsp yelahanka-sub-station, Description: This command will correct the ISIS LSP configuration and remove any incorrect LSP configuration or misconfigured ISIS protocol parameters that may be contributing to the ISIS LSP flooding.

## Post-Check Verification

```
Command: display bgp routing-table | include yelahanka-sub-station, Expected Output: BGP routing table details, including routing policies and filter lists., Description: This command will display the current BGP routing table, allowing us to verify that the BGP route filtering and routing policies have been corrected.

Command: display ospf neighbor | include yelahanka-sub-station, Expected Output: OSPF neighbor details, including interface configuration and protocol parameters., Description: This command will display the current OSPF neighbor details, allowing us to verify that the OSPF interface configuration and protocol parameters have been corrected.

Command: display isis lsp | include yelahanka-sub-station, Expected Output: ISIS LSP details, including LSP configuration and protocol parameters., Description: This command will display the current ISIS LSP details, allowing us to verify that the ISIS LSP configuration and protocol parameters have been corrected.
```

## Roll-back Procedures

- Step: Restore the original BGP configuration., Command: apply bgp router-id yelahanka-sub-station, Description: This command will restore the original BGP configuration, including any misconfigured route filtering or incorrect routing policies that may have been corrected.

- Step: Restore the original OSPF configuration., Command: apply ospf router-id yelahanka-sub-station, Description: This command will restore the original OSPF configuration, including any incorrect interface configuration or misconfigured OSPF protocol parameters that may

have been corrected.

- Step: Restore the original ISIS configuration., Command: apply isis lsp yelahanka-sub-station, Description: This command will restore the original ISIS configuration, including any incorrect LSP configuration or misconfigured ISIS protocol parameters that may have been corrected.